Wheeler–De Witt equation into a structure sufficiently well defined to be able to compute with. With this, it became possible to achieve precise results. The key technical result which clarifies the physical meaning of the graphs is the calculation of the spectra of volume and of area.

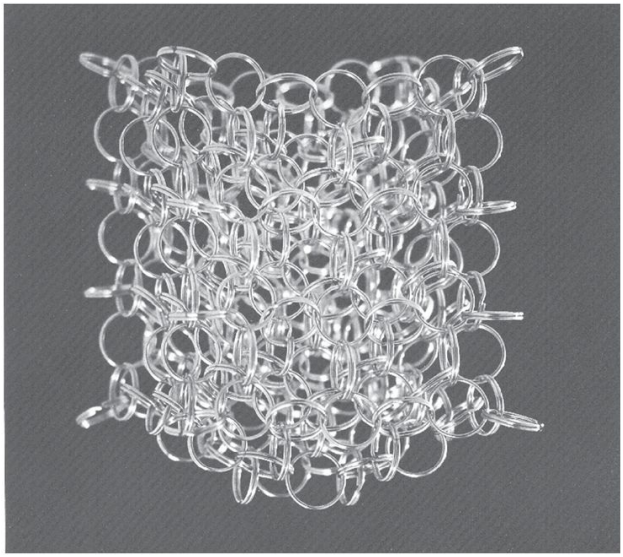


Figure 6.1 The quantum version of Faraday's lines of force, which weave space like a three-dimensional mesh of interlinked rings (loops).

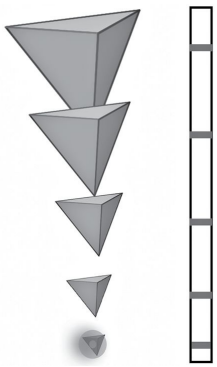


Figure 6.2 The spectrum of the volume: the volumes of a regular tetrahedron that are possible in nature are limited in number. The smallest, at the bottom, is the smallest volume in existence.

Spectra of volume and area

Take any region of space, for example, the room in which you are reading this, if you are in a room. How big is this room? The size of the space of the room is measured by its volume. Volume is a geometrical quantity which depends on the geometry of space, but the geometry of space – as Einstein understood, and as I recounted in Chapter 3 – is the gravitational field. Volume is therefore a property of the gravitational field, expressing how much gravitational field there is between the walls of the room. But the gravitational field is a physical quantity and, like all physical quantities, is subject to the